

BLACK RIVER FALLS AREA, WI, USA

WMO: 720859

Lat: 44.251N

Lon: 90.855W

Elev: 837

StdP: 14.26

Time Zone: -6.00 (NAC)

Period: 10-19

WBAN: 00286

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.6	-13.2	-27.9	1.2	-20.3	-20.4	1.8	-12.2	18.5	16.3	16.7	17.5	0.7	170	0.519

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.5	90.5	72.3	87.6	70.5	84.5	68.8	76.0	84.5	74.0	82.4	72.3	80.2	8.4	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.5	128.4	78.7	71.8	121.3	77.1	69.9	113.6	75.8	40.0	84.5	38.1	83.7	36.5	80.3	83.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.1	15.2	12.6	DB	-28.9	94.7	4.6	3.2	-32.1	97.0	-34.8	98.9	-37.4	100.7	-40.7	103.1	(4)
(5)			WB	-29.2	79.0	4.5	2.2	-32.4	80.6	-35.0	81.9	-37.5	83.1	-40.8	84.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	44.2	15.2	17.7	31.3	43.4	57.1	65.9	70.2	66.8	60.2	46.4	33.1	21.9		
	DBStd	21.69	14.05	13.70	13.13	9.09	9.09	6.38	5.88	5.84	8.89	8.89	10.37	12.29		
	HDD50	4423	1078	906	594	236	33	0	0	0	14	179	513	870		
	HDD65	8040	1543	1326	1044	650	278	64	18	46	199	579	958	1335		
	CDD50	2322	0	0	16	37	253	478	625	522	319	67	5	0		
	CDD65	462	0	0	1	0	32	92	178	103	54	2	0	0		
	CDH74	6102	0	0	13	40	562	1204	2285	1330	646	22	0	0		
	CDH80	1974	0	0	1	3	167	376	854	374	197	2	0	0		
(14) Wind	WSAvg	4.7	4.9	5.3	5.5	6.1	5.2	4.3	3.7	3.4	3.8	4.6	4.8	4.5		
(15) Precipitation	PrecAvg	35.7	1.2	1.0	1.9	3.3	4.2	5.5	4.3	4.8	3.7	2.7	1.7	1.3		
	PrecMax	43.9	2.9	2.0	3.4	6.6	10.6	10.3	10.5	10.7	10.9	6.3	3.2	2.6		
	PrecMin	25.9	0.4	0.4	0.5	1.1	1.6	2.1	1.0	1.0	0.8	0.9	0.1	0.4		
	PrecStd	4.2	0.6	0.5	0.9	1.8	1.9	2.5	2.3	2.5	2.3	1.5	0.9	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.1	57.2	74.6	79.4	90.5	91.9	96.6	91.5	90.8	76.7	66.8	51.3		
		MCWB	39.2	45.9	61.5	58.7	66.6	73.8	74.6	72.8	71.5	60.0	54.2	44.6		
	2%	DB	41.2	46.2	65.1	72.3	84.4	88.1	90.9	87.5	85.8	72.8	60.2	45.7		
		MCWB	37.4	40.0	54.0	53.8	63.4	69.3	74.3	70.7	69.3	59.4	51.5	42.1		
	5%	DB	37.0	41.1	58.5	67.4	80.3	84.3	87.7	83.9	81.3	68.3	54.6	41.7		
		MCWB	34.3	36.2	49.1	51.0	61.8	67.4	72.1	69.7	67.2	58.9	46.7	38.5		
	10%	DB	34.3	36.6	51.3	62.1	76.1	81.1	84.7	81.4	77.2	63.4	49.6	37.2		
		MCWB	32.7	32.7	44.6	48.0	60.9	66.3	69.8	67.7	65.9	54.9	44.6	35.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.1	46.7	62.5	61.6	71.2	76.9	80.2	77.0	75.1	69.2	57.4	47.9		
		MCDB	45.0	54.2	73.2	69.7	81.9	87.5	89.3	82.4	82.9	71.8	61.4	48.3		
	2%	WB	37.4	41.4	58.5	56.8	68.1	73.6	77.1	74.7	72.7	64.0	52.8	42.8		
		MCDB	40.7	45.7	64.3	67.4	77.1	82.4	85.5	81.7	79.9	67.4	60.9	43.7		
	5%	WB	35.1	36.5	49.6	54.0	66.1	71.4	74.8	72.8	70.8	60.1	47.8	38.0		
		MCDB	36.7	41.1	55.4	62.4	75.2	79.2	84.0	80.4	76.9	65.0	53.6	40.9		
	10%	WB	32.7	33.1	43.8	50.2	63.6	69.0	72.8	70.7	68.5	56.5	44.2	35.4		
		MCDB	34.3	35.9	50.4	61.1	71.5	75.6	81.1	77.8	74.8	62.4	48.6	37.2		
(35) Mean Daily Temperature Range	MDBR		20.7	23.0	23.5	25.3	26.6	25.7	26.5	26.5	27.1	23.7	19.7	16.9		
	5% DB	MCDBR	21.0	25.8	30.9	36.7	33.4	30.4	28.5	28.3	29.8	30.4	29.7	20.7		
		MCWBR	16.8	19.2	18.5	19.4	15.1	14.7	12.8	13.5	14.6	18.7	20.3	16.6		
	5% WB	MCDBR	19.6	23.9	24.4	28.8	25.9	23.7	24.4	23.0	22.8	22.7	26.1	17.3		
		MCWBR	16.0	18.5	15.6	17.2	12.8	13.3	12.4	12.4	13.2	17.5	19.0	14.4		
(40) Clear-Sky Solar Irradiance	taub		0.258	0.286	0.320	0.360	0.392	0.398	0.392	0.397	0.362	0.327	0.293	0.265		
	taud		2.404	2.323	2.383	2.345	2.300	2.318	2.345	2.332	2.424	2.508	2.518	2.455		
	Ebn at Noon		279	288	292	286	279	277	277	271	273	271	265	266		
	Edh at Noon		23	30	33	37	40	40	38	37	32	25	21	20		
(44) All-Sky Solar Radiation	RadAvg		560	912	1249	1433	1703	1882	1955	1654	1316	839	541	405		
	RadStd		61	86	146	141	132	147	91	110	94	106	49	45		

Historical Trends

		DBAvg		Heating		Cooling			Degree-Days					
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)			
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)		
(47)	Regional (52 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)		

Nomenclature: See separate page